

Linyi (Tiger) Hou

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EDUCATION

Ph.D. in Aerospace Engineering

May 2025

University of Illinois Urbana-Champaign

GPA: 3.94/4.00

Thesis: Autonomous Lost in Space and Time State Determination using Periodic Variable Stars

B.S. in Aerospace Engineering, Minor in Computer Engineering

May 2021

University of Illinois Urbana-Champaign

GPA: 3.93/4.00

SKILLS

Technical: batch filtering, Kalman filtering, Monte Carlo simulation, celestial navigation, orbit determination, spacecraft navigation modeling

Collaboration: technical/proposal writing, MS Office suite, $\text{\LaTeX}$, Git, GitHub, Trello

Programming: Python, C/C++, MATLAB/Simulink, GMAT, Javascript, Julia, ADQL

EXPERIENCE

Graduate Researcher, University of Illinois Urbana-Champaign

Jun 2021 – May 2025

- Developed an autonomous optical navigation concept using variable star light curves which delivers timing accuracy within 1 second without needing prior position/time information.
- Collaborated with colleagues to create navigation simulation tools in Python to assess the feasibility of reducing Deep Space Network tracking cadence by using onboard celestial navigation for the OSIRIS-APEX and CAPSTONE missions.
- Designed sensor scheduling algorithms for celestial navigation which reduced navigation uncertainty by over 10% compared to existing strategies in Monte Carlo simulation.
- Created a photon time of arrival simulation tool in Python and validated simulation results against the pulsar timing package PINT to within 50 μs .
- Developed a novel initial orbit determination technique using batched sequential range-rate measurements that achieved over 10x better accuracy than state-of-the-art methods.
- Developed a star identification technique using C++ which mitigates stellar aberration effects at relativistic speeds, improving star identification success rate from 38% to 91%.

Teaching Assistant, University of Illinois Urbana-Champaign

Jun 2021 – May 2024

- Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- Helped 20+ students formulate and implement UAV state-space model controllers with various sensors, e.g., laser rangefinders, optical flow cameras, and time-of-flight anchors.
- Assisted in developing LQR control-related code and instructional materials for a UAV course using C and Python.

Software Engineering Intern, Collins Aerospace

May 2019 – Aug 2019

- Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- Automated weekly builds and integration in TortoiseSVN using Python.

PROJECTS

Collaborative Projects in Flight Dynamics

2022, 2023

- Planetary entry: Created a 3-DOF entry simulation to assess the flight performance of several novel Apollo-derived entry guidance algorithms at Mars.
- Trajectory optimization: Applied optimal control theory to identify time- and fuel-optimal low-thrust Earth-Mars transfer trajectories.

Team Lead/Advisor, NASA's RASC-AL Competition

2019, 2022, 2023

- Led a team of 10+ students to develop a reusable crewed lunar lander concept (structure, power, thermal, life support, orbital mechanics, budget, etc.).
- Advised two student teams who developed technical requirements and performed trade studies for lunar and Martian in-situ resource utilization architectures.

HONORS AND AWARDS

<i>Aerospace Engineering Alumni Advisory Board Fellowship, UIUC</i>	2024
<i>Best Student Paper, 46th AAS GN&C Conference, American Astronautical Society</i>	2024
<i>Yee Memorial Fellowship, UIUC</i>	2021, 2024
<i>David and Catherine Thompson Space Technology Scholarship, AIAA</i>	2020

SELECT PUBLICATIONS

JOURNAL ARTICLES

- [J.3] **L. Hou**, Ishaan Bansal, Clark Davis, and Siegfried Eggl. "Position and Time Determination without Prior State Knowledge via Onboard Optical Observations of Delta Scuti Variable Stars". In: *The Journal of the Astronautical Sciences* 72.9 (2025). DOI: [10.1007/s40295-025-00485-8](https://doi.org/10.1007/s40295-025-00485-8).
- [J.2] **L. Hou** and Zachary R Putnam. "Autonomous Initial Orbit Determination using Sequential Range-Rate Measurements". In: *The Journal of the Astronautical Sciences* 71.24 (2024). DOI: [10.1007/s40295-024-00442-x](https://doi.org/10.1007/s40295-024-00442-x).
- [J.1] **L. Hou** and Zachary R Putnam. "A Norm-Minimization Algorithm for Solving the Lost-in-Space Problem with XNAV". In: *The Journal of the Astronautical Sciences* 71.6 (2024). DOI: [10.1007/s40295-023-00425-4](https://doi.org/10.1007/s40295-023-00425-4).

CONFERENCE PROCEEDINGS

- [C.3] **L. Hou** and Siegfried Eggl. "Autonomous Optical Navigation using Astrometry and Photometry". In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 25-248. Kaua'i, HI, Jan. 2025.
- [C.2] **L. Hou**, Ishaan Bansal, Clark Davis, and Siegfried Eggl. "Lost-in-space Position and Time Determination via Star Tracker Observations of Periodic Variable Stars". In: *46th Rocky Mountain AAS GN&C Conference*. AAS-24-012. Breckenridge, CO, Feb. 2024.
- [C.1] **L. Hou**, Kevin Lohan, and Zachary R Putnam. "Comparison and Error Modeling of Velocity-Based Initial Orbit Determination Algorithms". In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 21-280. Virtual, Feb. 2021.